

 De La Salle University	Research Ethics Review Committee Research Ethics Office, 3F Henry Sy Sr. Hall De La Salle University Manila 2401 Taft Avenue, Manila 1004, Philippines REO@dlsu.edu.ph (632) 524-4611 loc. 513	SOP No.: 2
		Form No.: 2.03
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DE LA SALLE UNIVERSITY

Checklist A

Research Ethics Checklist for Investigations involving Human Participants

*This checklist must be completed **AFTER** the De La Salle University Code of Research Ethics and Guide to Responsible Conduct of Research has been read and **BEFORE** gathering data. The University Code of Research Ethics is available at http://www.dlsu.edu.ph/offices/urco/forms/URCO-Code-of-Research-Ethics_August2011.pdf*

NOTE: This checklist is completed after the research proponent fills out the General Checklist Form.

Only answer this Checklist if you answered YES on question 1 of the General Checklist.

Researcher Details	
Students	BERNANDINO , Adrian Rafael DC. BORROMEIO , Anton Miguel G. GALAN , Russell Emmanuel G. GANAYO , Reinman Geoffrey A.
Thesis Adviser	DEL GALLEGGO , Neil
Department/College	Department of Software Technology / College of Computer Studies
Proposed Title of the Research	A New Rendering and Plume Simulation Framework for Visualizing Taal Volcano Eruption Plumes
Term(s) and academic year in which research project is to be undertaken	Term 3 of AY 25-26 to Term 3 of AY 26-27

Provide a brief description of the data collection procedure to be undertaken in the research:

Data collection consists of a two-pronged approach: a technical hardware benchmarking phase and a human usability testing phase. First, the research team will execute automated, software-driven rendering loops comparing the legacy OpenGL engine against experimental low-level explicit graphics API implementations (e.g., Vulkan or DirectX 12) under identical hardware environments to capture frames per second (FPS), frame render times, and CPU/GPU utilization logs. Second, targeted human usability tests will be conducted following a tiered recruitment strategy: PHIVOLCS personnel will be prioritized as the primary participant group given their domain expertise in volcanology; should this pool be

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insufficient, recruitment will expand, in order, to individuals who have witnessed a Taal Volcano eruption firsthand, then to volcano/geohazard simulation enthusiasts, and lastly to computer science students with a background in rendering or volcano-related coursework. Participants will interact with the multi-API 3D plume simulator configurations. After completing standard interaction tasks, participants will complete an anonymous, self-administered questionnaire using the System Usability Scale (SUS) to evaluate application responsiveness, stability, and visual parity.

The following should be attached to the checklist:

- A copy of the informed consent form to be used in the study.
- A copy of the instrument/tool that will be administered to the participants.
- If applicable, a copy of the letter seeking permission to collect data from participants who are under the supervision of an agency, institution, department, or office.
- If applicable, a copy of the parental consent form for participants below 18 years old.
- If applicable, a copy of the letter seeking permission to collect data from participants who are under the supervision of an agency, institution, department, or office

The following items refer to important ethical considerations in the conduct of research with human participants. Provide a check for the appropriate answer to each question.

Source of data

Please check all that apply:

☒	1. New data will be collected from human participants If you checked this item, how will the new data be gathered? Please check all that apply. After answering this question, please proceed to page 3
	☐ Experimental Procedures/Intervention/ Treatments
	☐ Focus Group
	☐ Personal Interviews
	☒ Self-administered Questionnaire
	☐ Researcher-administered Questionnaire
	☐ Internet survey
	☐ Observation
	☐ Telephone survey
	☐ Others, please specify:

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	2. Pre-existing data from human participants, i.e., from a dataset you checked this item, please proceed to page 7
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options are checked (both new data and pre-existing data), **answer all of the questions in this document.**

Only answer if new data will be collected (item 1 above)	
Sampling Details	
Number of Participants/Subjects	10 to 20 voluntary participants, recruited according to a tiered priority order: PHIVOLCS personnel as the primary target group, followed by individuals who witnessed a Taal Volcano eruption firsthand, then volcano/geohazard simulation enthusiasts, and lastly computer science students with a background in rendering or volcano-related coursework, should earlier priority groups be insufficient to reach the target sample size.
Location where the participants will be recruited/ where subjects will be obtained?	For PHIVOLCS personnel, recruitment will be initiated via email correspondence with the agency; if permission is granted, in-person coordination will also be done. For the remaining participant categories (Taal Volcano eruption witnesses, volcano/geohazard simulation enthusiasts, and computer science students or developers with a relevant background in 3D graphics or software development), recruitment will be conducted through a general online survey containing screening questions to confirm which category, if any, each respondent qualifies under.
How long will the data collection take place?	2 to 4 weeks during the system's evaluation timeframe.
Who will perform the data collection?	The primary student proponents (Bernardino, Borromeo, Galan, and Ganayo) under the adviser's supervision.
Location(s) where data collection will take place	DLSU GameLab, Gokongwei Building, Room 411 (G411).

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What procedures will be employed to ensure voluntary consent from participants?	Prior to the evaluation session, participants will receive a printed or digital Informed Consent Form (ICF) explicitly stating that participation is entirely voluntary, they may exit the software at any stage, and can choose to withdraw from the evaluation without academic or personal penalty.
Data Retention	
How long will data with participant identifiers be kept after the publication of the first paper from the project?	0 days (No personal identifiers will be collected from the beginning; all records remain fully anonymous).
How long will anonymized data be kept after the publication of the first paper from the project?	4 to 5 years following final publication or degree conferral to allow for reproducibility audits, securely saved in password-protected cloud environments before permanent deletion.
Procedure for Informed Consent	
How will informed consent be recorded? (check all that applies) Reminder: please attach informed consent that will be used in the study	☒ Written Consent ☐ Audio-recorded Consent ☒ Online/Email recorded Consent ☐ Others, please specify:

If you will not obtain a recorded informed consent, answer the questions that follow:

Why does the waiver of informed consent not pose a threat to the welfare and rights of the participants?

Why is recording an informed consent not practical for the proposed study?

	Yes	No	Not Applicable
1. Will the research involve students who will be receiving course credits for their participation?		✓	

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If YES, please attach a copy of the consent form and a summary of the debriefing process that will help participants understand how their participation in the research has provided a relevant learning experience to the crediting course.			
2. Does the study involve participants below 18 years old or those who are unable to give their informed consent? If YES, please attach a copy of the parental consent form.		✓	
3. Is there a possibility that the research can induce physical and/or psychological harm to the participants? Will they experience pain or some discomfort as a result from their participation in the research? If YES, please attach an acceptable argument that outlines the benefits of doing the research and how they outweigh the cost of harming the participants.		✓	
4. Will the participants be deliberately falsely informed or made unaware that they are being observed? Will they be misled in a way that they will possibly object to or show unease when told of the real purpose of the study? YES, please attach an acceptable argument that outlines the benefits of doing the research and how they outweigh the cost of harming the participants.		✓	
5. Will the research involve the discussion of, or questions on, sensitive topics (e.g. sexual activity, substance abuse, or mental health)? If YES, please make sure that the informed consent form explicitly states that sensitive questions will be posed and that you will safeguard the anonymity of the participants and ensure confidentiality. Please attach a copy of your informed consent form and your instrument.		✓	

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	Yes	No	Not Applicable
6. Will the research involve the administration of drugs, or other substances to the participants? YES, please attach an acceptable argument that outlines the benefits of doing the research and how they outweigh the cost of harming the participants. Please also attach a description of the procedure that will ensure that the participants will be brought back to their physical and psychological states prior to their participation in the research.		✓	
7. Will biological samples (e.g. blood, saliva, urine) be obtained from the participants? If YES, will this involve invasive procedures? Please attach a description of these procedures.		✓	
8. Will genetic materials be obtained from the biological samples? If YES, please attach a description of the procedures that will ensure confidentiality. Please attach the informed consent form.		✓	
9. Will financial inducements (other than reasonable expenses, like transportation or meal allowances) be offered to the participants for their participation in their research? If YES, the researcher(s) should be mindful of how the inducements can influence the participants' responses or behaviors during the research. Indicate the financial inducements offered to the participants: 		✓	
10. Is there a possibility for groups or communities to be harmed by the dissemination of the research findings?		✓	

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If YES, please attach a description of procedures to ensure the anonymity and confidentiality of the research findings.			
11. Will the results of this study have a commercial value? If yes, do you intend to apply for a patent for the output of this research? Please check: _____ Yes _____ No		✓	

FOR PROPONENTS WHO WILL GATHER NEW DATA ONLY, PLEASE STOP ANSWERING.

Use of Pre-existing Data collected from Human Participants		
Indicate the dataset from which the data for the study will be sourced		
Is the data publicly available, i.e., the access to which does not necessitate an approval process?	Yes	Please indicate where the dataset is available:
	No	Please indicate/attach the approval authority for access:
Was the original dataset originally collected for the present study's purpose?	Yes	Please attach the Consent Form used in the original study.
	No	Please attach the Information Collection Statement (i.e., the statement given to informants providing them with the rationale for the collection of specific information).
Does the original data set contain sensitive data, that is information that an individual would not likely want to be disclosed publicly, e.g., data on sexual activities, substance use?	Yes	Please describe the type of sensitive data to be used in the present research:
	No	
	No	

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Does the original dataset have personal identifiers?		<i>(This means that neither the researcher nor the participant provided any personal identifiers)</i>
		Yes, specifically: _____ Direct (i.e., the participant provided personal details like name and address) _____ Indirect (i.e., the participant was given a respondent code to make the participant identifiable)
Will new data be collected and analyzed along with data from the existing dataset?		Yes
		Please answer questions on page 3-5.
		No